

ORIGINAL RESEARCH

Superpixel segmentation algorithm based on local network modularity increment

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Abstract

Superpixel segmentation is a kind of image preprocessing technology and a popular research direction in image processing. The purpose of superpixel segmentation is to reduce the complexity of image processing. The most widely applied Simple Linear Iterative Clustering (SLIC) superpixel segmentation algorithm has high operating efficiency. However, under-segmentation is prone to occur when the number of given superpixel regions is too small. In order to improve the segmentation accuracy, the superpixel segmentation algorithm based on local network modularity increment (LocalNet) from the perspective of network community detection is proposed here. The adjacency network is constructed according to the colour similarity of image pixels, the local community centre is found by the degree of network nodes, and the local network structure of the community is constructed. The modularity increment is employed as the boundary constraint to improve the segmentation accuracy of superpixel segmentation. Through the experimental comparison with the SLIC algorithm, its improved algorithm, and the algorithm proposed in recent years, the results show that our LocalNet algorithm significantly improves in segmentation accuracy. Furthermore, the segmentation effect has obvious advantages under the premise that the segmentation speed is not much different from that of the other five algorithms.

1 | INTRODUCTION

The concept of superpixel [1] was proposed by Xiaofeng Ren in 2003. It divides the image into a fixed number of superpixel regions according to the consistency of the content. Superpixel is an image over-segmentation technology whose purpose is to use the image region as the processing unit instead of the image pixels, and reduce the complexity of image processing. The segmented image regions are connected and feature consistency, and still retain the local structure information of the image. Superpixel segmentation has been one of the hot research directions in image processing in recent years. It is used for image preprocessing and can significantly reduce the computational complexity of subsequent image processing tasks. Superpixel segmentation algorithms are widely applied in image analysis and computer vision fields, such as image/video segmentation

[2, 3], object detection [4, 5], sparse image representation [6], object tracking [7] etc.

So far, the representative superpixel segmentation algorithms are N-Cuts algorithm [8], SL (Superpixel Lattices) algorithm [9],

TurboPixels algorithm [10], and SLIC algorithm [11]. The N-Cuts is a segmentation method based on graph theory proposed by Shi J et al. in 2002. It quantifies the difference between the segmentation sub-images and the similarity within them as a segmentation quality evaluation function to segment the image iteratively. The superpixel segmentation results obtained by this algorithm are regular. However, the edge fit is low, and it is proved to be an NP-hard problem, which is complicated to solve and cannot handle larger images. Moore et al. [9] proposed the SL algorithm that required the edge information of the input image, and the edge cost is used as the evaluation function to find the optimal path in the horizontal and vertical

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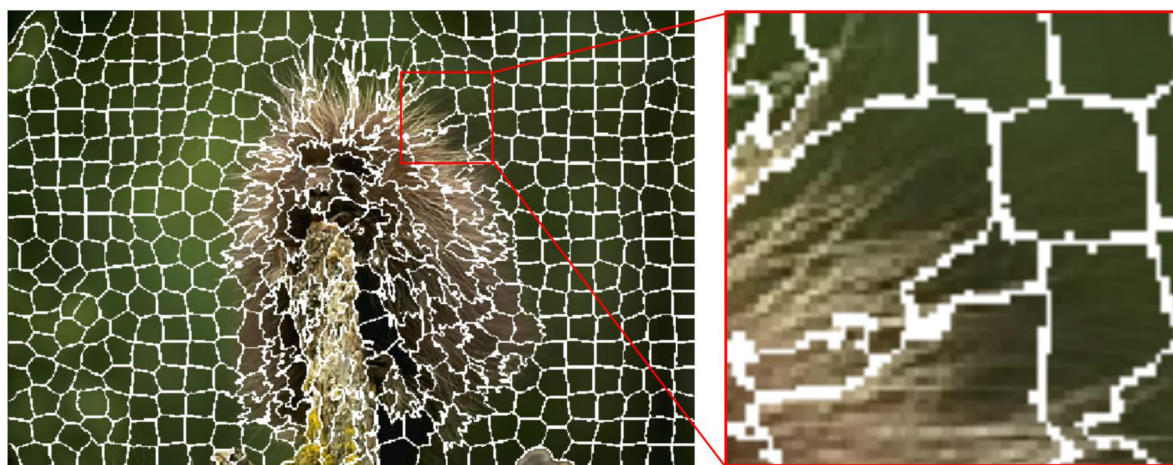


FIGURE 1 An example of under-segmentation of SLIC

directions for image segmentation to obtain the superpixel area of the image. This method relies too much on the initial input image edge information. Levinshtein et al. [10] proposed the level set TurboPixels algorithm based on geometric flow, which takes the non-boundary equidistant initial points as the origin, expands the origin according to the local gradient and boundary gradient changes of the image, and achieved the purpose of superpixel segmentation. The algorithm is also subject to under-segmentation by a ‘compactness constraint’, and the edge fit of the algorithm is still low. Achanta et al. [11] proposed the SLIC algorithm, which has accuracy excellent and low time complexity. However, the SLIC algorithm has no ‘compactness constraint’, and the segmentation results may be under-segmented when the number of given superpixels is small. Figure 1 shows an example of under-segmentation. In the red box of Figure 1, the SLIC algorithm divides the animal and background into the same superpixel area and resulting in under-segmentation.

In recent years, many scholars have made improvements to the shortcomings of the SLIC superpixel segmentation algorithm. For example, Psaltis et al. [12] extended the SLIC algorithm to a higher dimension to apply it to the field of hyperspectral image segmentation. They applied the ‘fractional distance’ to measure the distance between pixels to perform superpixel segmentation and improve the performance of hyperspectral image segmentation. The extended SLIC algorithm needs to select the maximum colour distance manually, and uses the same distance metric to different images, which leads to unsatisfactory segmentation results. In order to overcome the above problem, Kou F et al. [13] proposed a variance adaptive SLIC algorithm (VASLIC), which determined the ‘tightness factor’ of the algorithm according to the variance of the image neighbourhood and selected a more suitable ‘tightness number’ for each pixel. Each pixel adaptively determined the maximum colour distance and performed superpixel segmentation, which improved the segmentation accuracy of the algorithm. Zhou et al. [14] proposed the MMTDSLIC algorithm, which introduced the intermediary mathematical theory of dealing with blurring problems into the

superpixel segmentation problem and selected the maximum colour distance according to different images. This algorithm improved the segmentation accuracy to a certain extent. These improved algorithms for SLIC superpixel segmentation still do not restrict the boundaries of the image. When the number of superpixel regions is given small, under-segmentation is still prone to occur.

Li et al [15] proposed a CONIC algorithm, which introduced contour rating into the non-iterative clustering framework to improve conventional simple non-iterative clustering, to balance segmentation accuracy and visual consistency. Di et al. [16] proposed an ML-LISLIC algorithm, a hierarchical multi-level segmentation framework. Based on the simple linear iterative clustering algorithm, multiple discrimination and segmentation of the segmented image be added to increase the accuracy of superpixel segmentation.

In this paper, we introduce the network community detection into the superpixel algorithm to solve the problem of under-segmentation and improve the segmentation accuracy. The innovations of the method in this paper are as follows: (1) According to the characteristics of the similarity of the neighbours of the pixels in the image, the image local weighting network is constructed, and the community detection method is used in the constructed network to perform the image superpixel segmentation; (2) the modularity increment is employed as the segmentation boundary constraints to solve the problem of under-segmentation.

The rest of the paper is organized as follows. The related theory of community detection is described in Section 2. The construction of image local network and the LocalNet algorithm are introduced in Section 3. Experimental results are shown in Section 4. Conclusions are given in Section 5.

2 | RELATED THEORY

The qualitative and quantitative research on the characteristics of complex networks is called ‘network science’ [17]. Two ground-breaking papers have set off an upsurge in complex

network research. One is the small world network model published by Watts and Strogata in Nature [18], the other is the scale-free network model published by Barabás and Albert in Science [19]. They believe that most networks in the real world evolve dynamically, and the scale-free nature of actual networks comes from the network growth mechanism and the priority connection mechanism.

At present, scholars' research on complex networks mainly focuses on three aspects: studying and analysing the statistical characteristics of a large number of real networks; exploring the formation and internal mechanisms of networks and constructing network evolution models that conform to the statistical characteristics of real networks; applying the complex networks in specific fields.

The community structure of a complex network refers to a 'collection of nodes with a similar connection structure'. The nodes within the community are closely connected and sparsely connected with the nodes outside the community [20], which coincides with the purpose of superpixel segmentation of image. Therefore, community detection can be used for superpixel segmentation of image.

Newman's definition of community is that there are as many edges as possible within the community, and there are as few edges as possible between communities [21]. He proposed the concept of modularity to measure the quality of community division. The weighted network modularity is defined as [20]:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(l_i, l_j) \quad (1)$$

$$\delta(u, v) = \begin{cases} 1, & \text{if } u = v \\ 0, & \text{else} \end{cases} \quad (2)$$

where m is the sum of the weights of the edges in the network, and k_i is the sum of the weights of all the edges connected to the node i . A_{ij} represents the weight of the connected edge between nodes i and j . l_i is the label of the community where node i is involved in.

Louvain community detection algorithm found that if the current node i is an independent community, then the modularity increment when adding it to other communities j is calculated as [20]:

$$\begin{aligned} \Delta Q_i &= \left[\frac{\sum_{in} + k_{i,in}}{2m} - \left(\frac{\sum_{tot} + k_i}{2m} \right)^2 \right] \\ &\quad - \left[\frac{\sum_{in}}{2m} - \left(\frac{\sum_{tot}}{2m} \right)^2 - \left(\frac{k_i}{2m} \right)^2 \right] \\ &= \frac{k_{i,in}}{2m} - \frac{k_i \sum_{tot}}{2m^2} \end{aligned} \quad (3)$$

among them, $\sum_{in}$ is the sum of the weights of the edges connected in the community j , $\sum_{tot}$ is the sum of the weights of the edges connected to the community j (including internal and

external), k_i is the sum of the weights of the edges connected to the node i , and $k_{i,in}$ is the sum of the weights of the edges from node i connected to the community j .

The Louvain community detection algorithm is divided into three steps. In the beginning, each node in the network is regarded as an independent community, and the weight of the connected edges in the community is set to 0.

Step 1: Scan all nodes, traverse all neighbour nodes for each node, measure the modularity increment of adding the node to the community where the neighbour node is located, and select the neighbour node corresponding to the largest modularity increment to join the community. This process is repeated until the community ownership of each node no longer changes.

Step 2: Fold and merge the communities formed in Step 1. Each community is regarded as a single point and calculates the edge weights between the newly generated 'community points' and the sum of the join edge weights between all points in the community.

Step 3: Iteratively perform Steps 1 and Step 2 until the maximum modularity increment is less than zero, stop the iteration, and obtain the community division structure of the network.

3 | METHOD

In this section, we introduce the local network building method and the LocalNet superpixel segmentation algorithm.

3.1 | Construction of image local network

Each pixel of the image is regarded as a network node, and the entire image is constructed as a weighted network. Due to the large scale of the network, community detection takes a long time and takes up a large amount of storage space. To this end, a local weighted network of image is constructed in this paper.

In the beginning, for an image with M rows and N columns, each pixel is regarded as a node of the network. Then calculate the colour similarity S_{ij} between each node i and its 8-neighbor node j :

$$S_{ij} = \exp \left(\frac{-D(c_i, c_j)}{2\sigma^2} \right), \quad i = 1, 2, \dots, M \times N, \quad j \in U(i) \quad (4)$$

$$D(c_i, c_j) = (c_i - c_j)^T (c_i - c_j) \quad (5)$$

where $U(i)$ represents the 8-neighbor of node i , $\sigma > 0$ is a constant, and c_i represents the colour feature vector of node i . Each node only has edges between its 8-neighbor nodes, and the weight of the edge is S_{ij} . At this time, a weighted network of all nodes of the image is constructed. The degree d_i of node i is defined as the sum of the weights of the edges connected to node i [21], that is:

$$d_i = \sum_{j \in U(i)} S_{ij} \quad (6)$$

ALGORITHM 1 Local Network

Input: an image I with M rows and N columns, the number K of superpixel regions

Output: the label map L of the image I

/*Initialization*/

Calculate the colour similarity S_{ij} between each node i and its 8-neighbor node j , and the degree d_i of node i .

while not all nodes are in the community **do**

Find the node i^* with maximum degree that is not belongs to any communities as centre of the current local network N_{i^*} . Calculate the modularity increment ΔC_{i^*} when the nodes joining community C_{i^*} .

if $\Delta C_{i^*} \geq 0$ and the numbers of C_{i^*} is less than MN/K **then**

Join the nodes to the C_{i^*}

update N_{i^*} , C_{i^*} , Calculate the modularity increment ΔC_{i^*} when the nodes joining community C_{i^*} .

end if

end while

Merge communities with the number of nodes smaller than $MN/10K$.

Assign the same label to all nodes within the community.

Return the label map L of the image I .

When the degree of the node is larger, the node's importance is higher, and the similarity between the node and its surrounding area's nodes is much greater.

3.2 | LocalNet algorithm

A pseudo code of the LocalNet algorithm is presented in Algorithm 1. The specific algorithm flow is as follows.

For image I with M rows and N columns, superpixel division is performed, and the number of superpixel regions is set as K .

Step 1: Construct the adjacency weighting network of the image, each pixel of the image is regarded as a network node, according to the image RGB colour vector, combined with formulas (4) and (5) to calculate the similarity weight of each node and its 8-neighbor nodes which regard as the node connection weight, the degree of the image adjacent to the weighted network node is calculated according to formula (6).

Step 2: In the construction of the image local weighted network, the node with the largest degree that has not been merged into the community is selected as the current community centre.

Step 3 Calculate the modularity increment when the node that has 8-adjacency with the current community node joins the community structure and find the node corresponding to the modularity increment greater than zero to merge into the current community.

Figure 2 shows the local network community detection diagram. When a node (marked in red) is selected as the current community, the nodes that need to calculate the modularity increment include only the 8-adjacent nodes around the community. When the community contains three nodes (marked in red), the modularity increment needs to be considered when calculating the modularity increment of 12 nodes around it, and so on.

Step 4: Repeat Step 3 until the modularity increment of all nodes in the local weighted network joining the community is less than or equal to zero, or the number of nodes in the

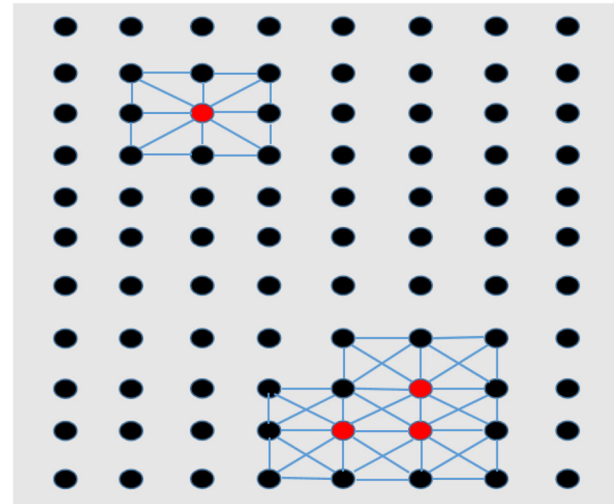


FIGURE 2 Local network community detection diagram of image

community is equal to $\frac{MN}{K}$, then the community expansion is stopped.

Step 5: Repeat Step 2 to Step 4 until all nodes in image network are allotted into communities.

Step 6: The communities whose number of nodes are less than $\frac{MN}{10K}$ are merged into their neighbouring larger communities according to the nearest neighbour algorithm.

4 | EXPERIMENTS

4.1 | Analysis of segmentation accuracy

In order to verify the effectiveness of the LocalNet superpixel segmentation algorithm, comparative experiments are conducted on the natural image segmentation and benchmark dataset [22] BSDS500, and the dataset ADE20K_2017 [23]. The BSDS500 dataset contains 500 natural images, which the Berkeley computer vision group provides for segmentation or contour detection. The ADE20K_2017 is a scene parsing benchmark which contains 150 object and stuff classes. The SLIC algorithm [11], VASLIC algorithm [13], MMTDSLIC algorithm [14], CONIC algorithm [15], ML-LISLIC algorithm [16], and our algorithm (LoclaNet) are compared experimentally. The partial segmentation results of each algorithm are shown in Figures 3–5.

Figures 3 and 4 show the segmentation results of the SLIC algorithm and the LocalNet algorithm when the number of superpixel regions is set to 500 and 2000, respectively. The red box is an enlarged view of the superpixel segmentation details. Comparing Figures 3 and 4, it can be seen that the SLIC algorithm segmentation results are much regular. The LocalNet algorithm is significantly better than the SLIC algorithm in segmenting the finer edge parts.

Figure 5a shows the segmentation results of the SLIC algorithm when the number of superpixel regions is set to 300. It can

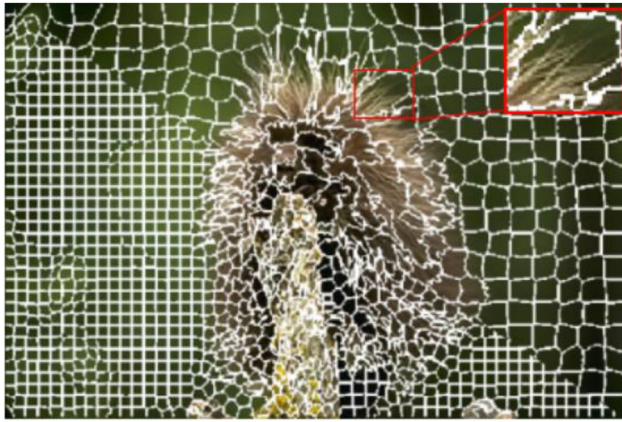


FIGURE 3 SLIC superpixel segmentation

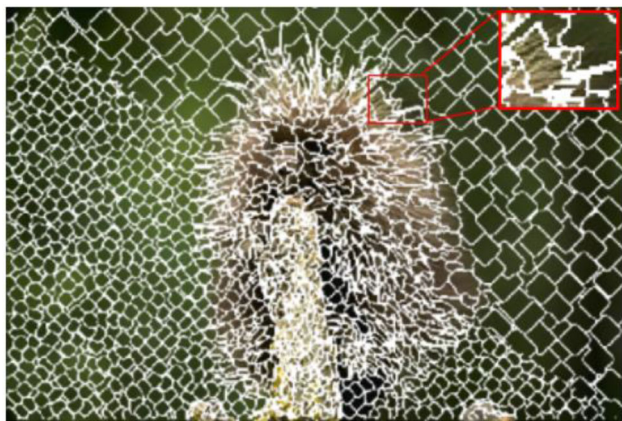


FIGURE 4 Local Net superpixel segmentation

be seen that for most image regions, the algorithm has a better segmentation effect and can be segmented along the edge of the object. However, for the details, such as the enlarged part in the red box and the eagle's shoulder in Figure 5a, the SLIC algorithm does not separate the eagle from the sky, resulting in under-segmentation. Figure 5b shows the segmentation results of the MMTDSLIC algorithm. Compared with the SLIC algorithm, the segmentation results are much regular, but the resulting under-segmentation error is quite large. Figure 5c shows the segmentation result of the VASLIC algorithm. Compared with the SLIC algorithm, there is a certain improvement. The shoulder of the eagle has been separated from the sky, but the enlarged part in the red box is still not distinguished. Figure 5d shows the segmentation results of the CONIC algorithm. It can be seen that compared with the above three algorithms, some objects and backgrounds have been separated, but there is still under-segmentation. Figure 5e shows that the ML-LISLIC algorithm has separated the background and object of the red box area. However, it can see intuitively that the segmentation is very irregular, and some segmentation errors occur. Figure 5f shows that the segmentation results of the LocalNet algorithm in this paper have completely separated the eagle and the sky on the shoulder of the eagle show in the red magnified box, and the

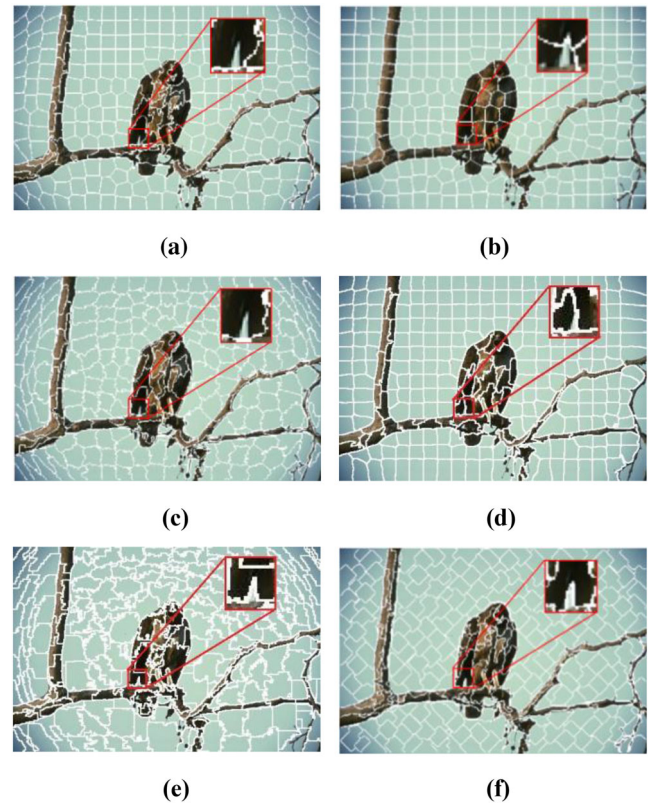


FIGURE 5 Comparison of the segmentation results of the six algorithms. (a) SLIC (b) MMTDSLIC (c) VASLIC (d) CNOIC (e) ML-LISLIC (f) LocalNet (The number of super pixels is 300)

segmentation effect is significantly better than the other algorithms above.

The standard boundary recall (BR) and under-segmentation error (USE) are used to measure the edge segmentation of the algorithm [11].

1. Boundary recall

$$BR = \frac{TP}{TP + FN} \quad (7)$$

Among them, TP is the number of boundaries that divide the correct boundary (the boundary in Ground Truth) into boundaries, and FN is the number of boundaries that divide the correct boundary into non-boundaries. The larger the BR value, the closer the boundary obtained by superpixel segmentation to the boundary in Ground Truth, and the better the superpixel segmentation effect.

2. Under-segmentation error

The under-segmentation error is used to measure the difference between the results of image segmentation algorithm and the ground truth.

The ground truth of an image divides the image into regions $\{g_1, \dots, g_k\}$ and the superpixel algorithm divides the image into regions $\{s_1, \dots, s_L\}$. The USE of the segmentation result is

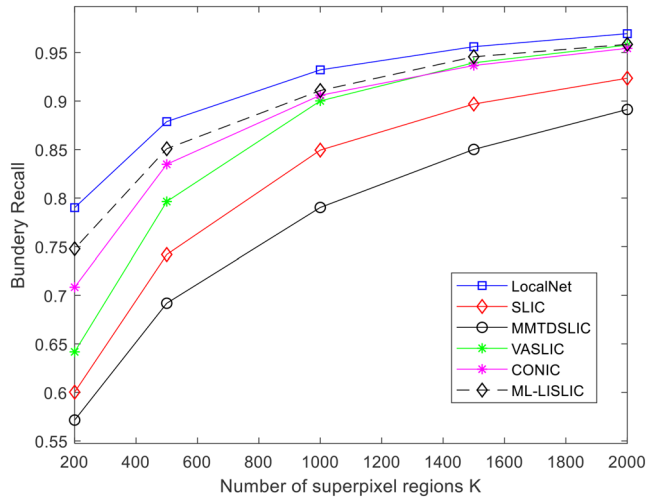


FIGURE 6 The BR of each algorithm in BSD500

TABLE 1 The BR of each algorithm in BSD500

Algorithm ↑	Number of superpixel regions				
	200	500	1000	1500	2000
SLIC	0.6003	0.7419	0.8494	0.8969	0.9235
MMTDSLIC	0.5715	0.6919	0.7903	0.8501	0.8913
VASLIC	0.6417	0.7964	0.9001	0.9394	0.9576
CONIC	0.7084	0.8348	0.9059	0.9366	0.9543
ML-LISLIC	0.7478	0.8508	0.9108	0.9456	0.9584
LocalNet	0.7901	0.8788	0.9321	0.9560	0.9694

Formula (8), where $Area(s_j)$ represents the area of the segmented region s_j [11].

$$USE = \sum_{i=1}^k \frac{[\sum_{\{s_j | s_j \cap g_i \neq \emptyset\}} Area(s_j)] - Area(g_i)}{Area(g_i)} \quad (8)$$

The larger the USE value, the greater the superpixel segmentation error, so the smaller the USE, the better the superpixel segmentation effect.

On the datasets BSDS500 and ADE20K_2017, the LocalNet algorithm in this paper is compared with the SLIC algorithm, the VASLIC algorithm, the MMTDSLIC algorithm, the CONIC algorithm and the ML-LISLIC algorithm. The number K of superpixel regions is set to 200, 500, 1000, 1500, 2000, respectively. The mean of BR and USE are calculated for 500 images from the BSDS500, 2000 images from the Validation set in ADE20K_2017, respectively. The experimental results are shown in Figures 6–10 and Tables 1–5. According to Figures 6, 9 and Tables 1, 4, it can be seen that the BR of the MMTDSLIC algorithm is poor, and the VASLIC algorithm has improved compared with the SLIC algorithm. Both the CONIC algorithm and the ML-LISLIC algorithm are improved. The BR of the LocalNet algorithm in this paper is significantly higher than that

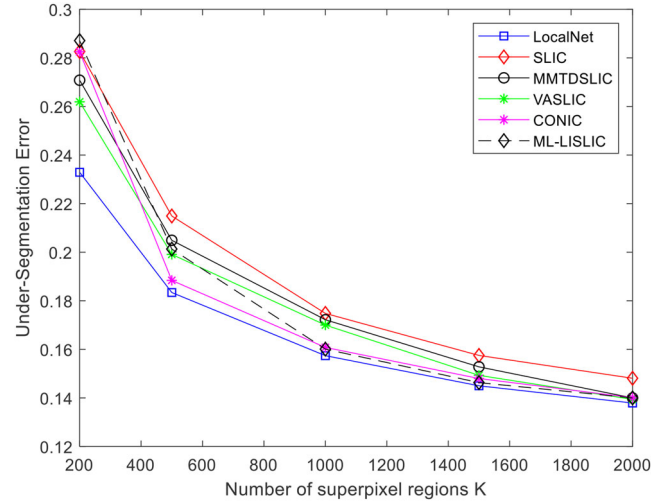


FIGURE 7 The USE of each algorithm in BSD500

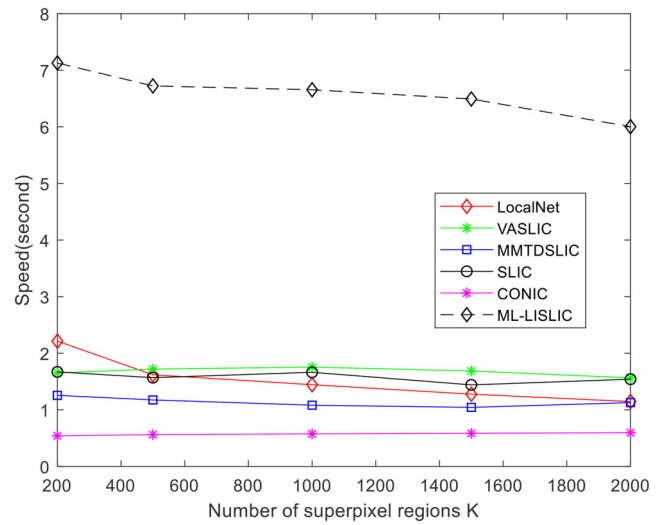


FIGURE 8 Comparison of speed of each algorithm (second)

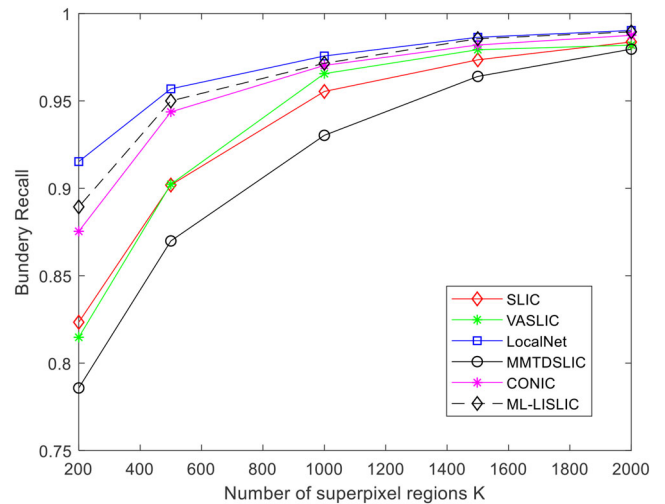


FIGURE 9 The BR of each algorithm in ADE20K_2017

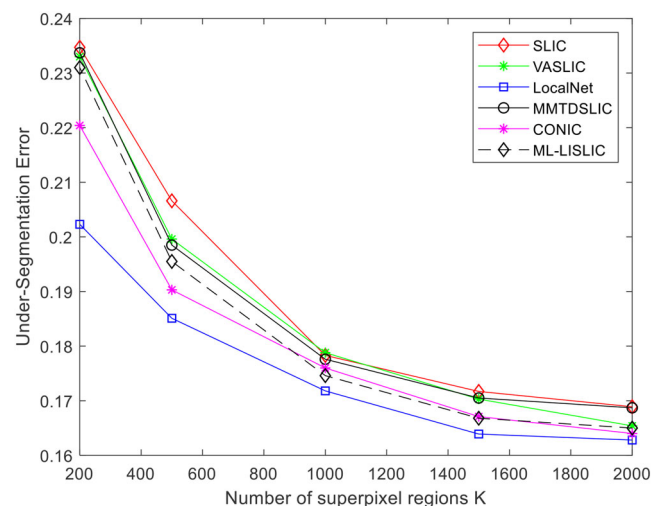


FIGURE 10 The USE of each algorithm in ADE20K_2017

of other algorithms, indicating that the boundary of our algorithm is much accurate. Figures 7, 10 and Tables 2, 5 show that compared with the SLIC algorithm, the MMTDSLIC algorithm has a smaller USE. When the number of superpixel regions is small, the USE of the MMTDSLIC algorithm and the VASLIC algorithm is basically the same, and as the number of superpixel regions increases, compared with the MMTDSLIC algorithm and the SLIC algorithm, the USE of the CONIC algorithm and the ML-LISLIC algorithm is significantly reduced. The LocalNet algorithm in this paper is better than the other algorithms above when the number of superpixel regions is from small to large. In summary, the superpixel segmentation algorithm based on local network modularity increment proposed in this paper is effective.

4.2 | Computational complexity analysis

The SLIC algorithm uses linear iterative clustering to determine the category of pixels. Each clustering complexity of the algorithm is close to $O(1)$, and the algorithm runs very efficiently. The MMTDSLIC algorithm uses fuzzy mathematics theory to select the maximum colour distance, cancels the iterative steps of the SLIC algorithm, and improves the operating efficiency compared to the SLIC algorithm. The VASLIC algorithm needs to calculate colour variance every iteration, so the operation efficiency is slower than the SLIC's and MMTDSLIC's, but the segmentation effect is much better. The LocalNet algorithm in this paper is different from the SLIC algorithm and its improved algorithm. The application of complex network theory for superpixel segmentation has the advantage of being able to better deal with the problem of unconstrained image segmentation boundaries and not prone to under-segmentation. However, the amount of calculation is relatively large. The LocalNet algorithm is accelerated by processing the image in parallel in regions.

Table 3 shows the running time comparison results of the SLIC algorithm, the MMTDSLIC algorithm, the VASLIC algo-

TABLE 2 The USE of each algorithm in BSD500

Algorithm ↓	Number of superpixel regions				
	200	500	1000	1500	2000
SLIC	0.2826	0.2149	0.1748	0.1575	0.1481
MMTDSLIC	0.2708	0.2049	0.1722	0.1528	0.1400
VASLIC	0.2619	0.1991	0.1700	0.1493	0.1394
CONIC	0.2825	0.1884	0.1608	0.1480	0.1401
ML-LISLIC	0.2871	0.2031	0.1601	0.1436	0.1400
LocalNet	0.2329	0.1834	0.1574	0.1450	0.1379

TABLE 3 Speed comparison of various algorithms (second)

Algorithm ↓	Number of superpixel regions				
	200	500	1000	1500	2000
SLIC	1.6704	1.5686	1.6636	1.4408	1.5414
MMTDSLIC	1.2562	1.1750	1.0820	1.0436	1.1278
VASLIC	1.6518	1.7202	1.7558	1.6862	1.5646
CONIC	0.5411	0.5621	0.5759	0.5854	0.5960
ML-LISLIC	7.1280	6.7228	6.6550	6.4916	6.0021
LocalNet	2.2140	1.6194	1.4442	1.2770	1.1462

TABLE 4 The BR of each algorithm in ADE20K_2017

Algorithm ↑	Number of superpixel regions				
	200	500	1000	1500	2000
SLIC	0.8234	0.9019	0.9554	0.9735	0.9837
MMTDSLIC	0.7858	0.8699	0.9303	0.9640	0.9796
VASLIC	0.8148	0.9023	0.9656	0.9793	0.9819
CONIC	0.8754	0.9437	0.9703	0.9820	0.9875
ML-LISLIC	0.8894	0.9500	0.9715	0.9856	0.9895
LocalNet	0.9153	0.9569	0.9757	0.9864	0.9903

gorithm, the CONIC algorithm, the ML-LISLIC algorithm and the LocalNet algorithm. The running time in Table 3 are the average consumption time of 500 images from the BSDS500 dataset when the number of superpixel division regions set as 200, 500, 1000, 1500 and 2000, respectively. According to Figure 8 and Table 3, the running speed of the CONIC algorithm is significantly faster than the other algorithms. The LocalNet algorithm is slightly inferior to the other algorithms when the number of superpixel regions is small. When the number of superpixel regions is large, the computing efficiency of the LocalNet algorithm is slightly different from the fastest algorithm. However, the segmentation accuracy is significantly better than the other five algorithms.

TABLE 5 The USE of each algorithm in ADE20K_2017

Algorithm ↓	Number of superpixel regions				
	200	500	1000	1500	2000
SLIC	0.2347	0.2066	0.1783	0.1717	0.1689
MMTDSLIC	0.2337	0.1985	0.1776	0.1705	0.1687
VASLIC	0.2330	0.1996	0.1788	0.1703	0.1654
CONIC	0.2204	0.1903	0.1760	0.1671	0.1640
ML-LISLIC	0.2310	0.1955	0.1746	0.1668	0.1650
LocalNet	0.2023	0.1851	0.1718	0.1639	0.1628

5 | CONCLUSION

As an image preprocessing technology, the superpixel segmentation algorithm has been widely applied in the field of image segmentation and object detection. Superpixel segmentation can also be applied to image feature extraction and region-based image analysis with high requirements for details. Superpixel regions can be used to replace pixels as image processing units, which dramatically reduces the memory and time consumption of image processing. In order to solve the problem of under-segmentation in image segmentation when the number of superpixel regions is set small (generally less than 500), a superpixel segmentation algorithm based on local network modularity increment, called the LocalNet algorithm, is proposed in this paper. The constructed image local network better describes the characteristics of the image. The modularity increment is introduced as the boundary constraint condition of segmentation, so that the selection of community detection nodes is only selected in the local network, which is more in line with the distribution characteristics of the image, reduces the complexity of community detection and improves the segmentation accuracy of the algorithm. The experimental results show that the LocalNet algorithm proposed in this paper, compared with the SLIC algorithm and its improved algorithms VASLIC, MMTDSLIC and the latest CONIC, ML-LISLIC algorithms has a significant improvement in segmentation accuracy when the segmentation speed is not much different from the other algorithms and has satisfactory segmentation performance. However, the LocalNet algorithm requires the additional calculation of node modularity and other evaluation criteria to measure node division, so its speed still needs to be improved.

ACKNOWLEDGEMENTS

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CONFLICT OF INTEREST

None

DATA AVAILABILITY STATEMENT

BSDS500 dataset http://www.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/BSR/BSR_bsds500.tgz

ADE20K_2017 dataset: <http://groups.csail.mit.edu/vision/datasets/ADE20K>

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